

Assignment 6

Objective: Compare the performance, decision-making, and complexity of Logistic Regression, Support Vector Machines (SVM), and Neural Networks (NN).

Dataset: [Pima Indians Diabetes Database](#)

1. Experimental Design

You are responsible for the end-to-end data pipeline. You must decide and justify:

- **Data Preparation:** How you handle missing values or anomalies.
- **Feature Engineering:** Which features to keep, transform, or combine.
- **Train/Test Split:** Use the first two digits of your roll number as the `random_state` for your split and all model parameters. (If your roll number is 22L-45607, use 45).

2. Model Implementation

Implement and tune the following four models. Aim to optimize hyperparameters for the best possible performance.

1. Logistic Regression
2. SVM (Linear Kernel)
3. SVM (RBF Kernel)
4. Neural Network (MLP):
 - Must use at least two hidden layers (e.g., 16 and 8 neurons).
 - Use the ReLU activation function for hidden layers.

3. Visual Comparison

Visualize the following:

- **Correlation Heatmap:** To show which features are most strongly linked to the target.
- **Performance Bar Chart:** Comparing Accuracy, Precision, Recall, and F1-Score for all models.
- **Combined ROC Curve:** Plot all four models on a single graph and report the AUC.
- **Training Time:** A chart comparing the time taken to train each model.

4. Submission

Submit your work as a single Jupyter Notebook (.ipynb). Your notebook should be structured as follows:

- **Methodology:** Clear cells explaining your data choices and pipeline logic.
- **Code:** Well-commented Python code implementing the preprocessing, models, and visualizations.
- **Results:** The required visuals followed by a brief summary of your findings.